

THOMAS REESE

Software Engineer · Full-Stack & AI Systems

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SUMMARY

Full-stack engineer (4+ yrs) shipping AI-native systems on AWS Bedrock and the Model Context Protocol. Currently at Envoy building Codebase Intelligence: an 87,500-chunk vector index serving 22 repos at ~\$0.03/month. Production stack spans Ruby on Rails, Ember.js, React 19, and Python; comfortable from infra (Pulumi/Terraform) to model eval (RAG, MCP tool use, embeddings).

EXPERIENCE

Envoy | Software Engineer, Visitor Management

Remote · Nov 2024 – Present

- Designed and shipped **Envoy's Codebase Intelligence platform** end-to-end across four phases: **Amazon S3 Vectors** infrastructure on AWS (Pulumi/Terraform), an **Amazon Bedrock Knowledge Bases** store using **Titan Text Embeddings V2**, and a Python **MCP server** exposing six retrieval tools to **Anthropic Claude**. Indexed 22 internal repositories (~87,500 **Tree-sitter** AST chunks) with top-1 retrieval averaging **0.785** at ~\$0.03/month, replacing per-engineer 6–8 GB local vector databases.
- Drove the migration of the visitor-management dashboard from **Ember.js** / **Glimmer** to **React 19** across the visitor profile, printer overview, and new screeners, visits, matching-rules, and documents tabs, using **Module Federation 2.0**, **TanStack Query v5**, **Radix UI**, **Tailwind**, **Luxon**, and **MSW v2**.
- Shipped the **Visitor Profile Merge** feature end-to-end across ~20 tickets (visitor directory, merge UI, Rails APIs) and the **Block List V2** identity-matching lifecycle (expiration dates, disable toggle, search, deletion) across two quarters.
- Owned the frontend for a ChargeBee-feature-code-driven subscription gate on Envoy's Visitors Basic plan, replacing fragile SKU-name matching with a counter table, gate-state API, 75% / 100% threshold mailer, and reset cron. Shipped across six PRs spanning Rails and Ember in a single sprint.
- Scoped Envoy's Visitors AI roadmap of ~30 child tickets (AI drafting, fuzzy matching, NL analytics, audit log, smart dedup) and the **AgentLock** initiative defining phase-based agent permissions for the 14 AI tools used across the Visitors product.
- Resolved ~20 customer-facing bug tickets affecting Fortune 500 accounts including **Blue Origin**, **Tesla**, and **NVIDIA**, using TDD with **RSpec** on the Rails side and **Vitest** + **MSW** on the React side, with mutation-tested verification against staging via Kubernetes pod exec.
- Authored **28+ internal technical documents** adopted cross-team, including an AI-tooling onboarding guide, a six-layer AI memory architecture reference, an AI coding-agent observability writeup, and the Engineering Innovation Program's cross-team companion guide.
- Sustained **337 merged pull requests** across nine internal repositories in 18 months on a stacked-PR (**Graphite**) and worktree-based workflow, shipping Ruby on Rails, Ember Octane / Glimmer / GTS, React 19, **Cerbos** authorization, and Pulumi / Terraform IAM changes at a 5+ PR/week cadence without rollback.

Tech: Ruby on Rails · Ember.js (Octane / Glimmer / GTS) · React 19 · TypeScript · PostgreSQL · Redis · Kafka · Sidekiq · Cerbos · TanStack Query · Module Federation · Tailwind · Radix UI · MSW · Vitest · RSpec · Playwright · AWS (Bedrock, S3 Vectors, IAM) · Kubernetes · Pulumi · Terraform · OpenTelemetry · Datadog · Graphite · Anthropic Claude / Model Context Protocol

Sprout Social | Front-End Engineer

Chicago, IL · May 2022 – April 2024

- Drove the front-end migration of key components from **Flow** to **TypeScript**, improving code quality through stricter type checking and cutting daily-workflow debugging time by ~20%.
- Built and maintained reusable **React** components in the **Seeds** shared library, standardizing UI patterns across teams and reducing feature development time by 3+ days per project.
- Partnered with product managers and designers to integrate UX/UI improvements across the platform. Conducted code reviews and mentored junior developers on best practices.
- Built a React UI for bulk sentiment reclassification, allowing users to reclassify up to **1,000 messages** per batch with Redux state and Axios, cutting per-batch handle time ~30%. Optimized rendering with React memoization and virtualized lists.
- Developed dynamic **Query Builder Form** components in React + TypeScript for AI-driven keyword suggestions using **OpenAI's GPT** model, applying debounce, async/await, and React Hooks to reduce query setup time.
- Created a faceted message-exploration tool with **React Context API**, Redux, and React Router, using SCSS for responsive styling and virtualized lists for performance, increasing user engagement ~30%.

The Walt Disney Company | Software Engineering Intern

Remote · May 2021 – Nov 2021

- Improved site quality across multiple **ABC news stations**: fixed broken links, restructured navigation, and addressed responsive UI/UX issues using React and CSS Flexbox/Grid.
- Developed a dynamic **Banner-Maker** tool for ESPN: reusable React banner components with Redux-managed state and team/player API integration, streamlining banner creation workflows by ~35%.
- Owned front-end design and implementation for a Student Loan Refinance project: responsive React UI, input validation, and API integration with the back-end team, reducing form validation errors by ~60%.
- Partnered with **Applitools** stakeholders on accessibility: screen-reader testing and WCAG compliance, boosting accessibility coverage by ~70%.

Credit Karma | Full-Stack Web Developer Intern

San Francisco, CA · Jan 2020 – Dec 2020

- Contributed to internal full-stack tooling across **React**, **Node.js**, **Ruby**, and **Python** services on AWS, with a focus on infrastructure automation and front-end developer experience.
- Built and reviewed pull requests across multiple repositories, partnering with engineers and QA on test coverage and code quality.
- Featured as the credit-recovery story in a **Credit Karma** brand commercial focused on personal finance impact.

SKILLS

<i>Languages</i>	TypeScript, JavaScript (ES6+), Python, Ruby, HTML5, CSS3, SQL
<i>Frameworks & UI</i>	React 19, Next.js, Ember.js (Octane / Glimmer / GTS), Ruby on Rails, FastAPI, Node.js, Express, Tailwind, Radix UI, SCSS, Three.js, Framer Motion
<i>AI & Data</i>	Model Context Protocol (MCP), Anthropic Claude SDK, OpenAI SDK, Amazon Bedrock, S3 Vectors, Titan Text Embeddings V2, Tree-sitter AST chunking, ChromaDB, Pinecone, LangChain
<i>Cloud & Infra</i>	AWS (Bedrock, S3, IAM), Pulumi, Terraform, Kubernetes, Docker, OpenTelemetry, Datadog, Vercel
<i>Data Stores</i>	PostgreSQL, Redis, Kafka, Sidekiq, MongoDB
<i>Build & State</i>	Module Federation 2.0, Vite, Webpack 5, TanStack Query v5, Redux Toolkit, MSW v2, GraphQL, REST
<i>Testing & CI</i>	Vitest, RSpec, Playwright, Jest, Cypress, Graphite (stacked PRs), GitHub Actions, CircleCI
<i>AuthZ & Security</i>	Cerbos, OAuth, JWT, WCAG accessibility

EDUCATION

Dominican University

B.S. Computer Science

River Forest, IL

Aug 2018 – May 2022

PROJECTS

TraderDan — Automated Trading System [\[github\]](#)

Python · Next.js · FastAPI · Anthropic Claude · TradeLocker · PostgreSQL · Optuna · ChromaDB

Production trading system managing real capital across five active prop-firm accounts, trading XAUUSD on H1/M15 via TradeLocker. Architecture spans a backtest engine, live signal generator, FastAPI webhook execution server with prop-firm risk gates (daily-loss, drawdown, ATR-spike, news blackout, weekend closer), a 14-agent AI team led by an OpenClaw CEO orchestrator (Anthropic Claude tool use + Gemini Flash routing), an eval harness with drift detection, semantic memory in ChromaDB, and a Next.js 15 dashboard with 17 pages for live monitoring. 5,696+ tests on a stacked-PR workflow.

Blue Ridge Canine Academy [\[link\]](#)

React · Node.js · Express · MongoDB · Acuity Scheduling

Freelance build (\$800) for a local dog trainer: a full-stack scheduling and payments platform. React frontend, Node.js / Express backend, MongoDB persistence. Wireframed in Figma and hosted on GoDaddy.

HONORS & AWARDS

- Won **1st place** at the Google × Code Nation Hackathon, guiding local students in HTML, CSS, and JavaScript to build passion projects.
- Won the **24-hour Code for Good Hackathon** by JP Morgan, leading a front-end team that shipped a dynamic site for the Girls in the Game non-profit.
- Won **2nd place** at the ACCA Programming Contest (Associated Colleges of the Chicago Area), a record-setting finish for Dominican University.
- Completed the **MLT Career Program**, sharpening leadership, networking, and communication skills.
- Mentored high school students in STEM and technology at **Code Nation**.
- Featured in a **Credit Karma** commercial on personal credit recovery; also appeared in a **JumpStart** rebranding spot on early-career recruiting.